

Risk-Aware Highway Lane-Change Planning via Online Soft-Constrained CVaR-MPPI

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Abstract—Standard Model Predictive Path Integral (MPPI) control is a sampling-based, online dynamic programming method that approximates the stochastic Hamilton-Jacobi-Bellman value function via path integrals. It is risk-neutral by construction. For highway lane changes the importance-weighted mean hides the tail: a 1-in-256 collision is invisible. This project formulates a CVaR-constrained lane-change problem and solves it with an online CVaR-augmented MPPI controller. The originally proposed hybrid method (offline CVaR-MDP value iteration supplying the terminal cost, online MPPI executing the receding-horizon plan) was descope to the online level only, for the reasons stated in §I-B. The deployed controller samples 256 control sequences per step, evaluates each candidate’s worst $1-\alpha$ tail over 64 disturbed traffic futures, and adds a gated tail-cost penalty to the rollout cost. Evaluated on a HighwayEnv simulator across three scenarios (lane change, cut-in, hard-brake leader), CVaR-MPPI prevents a collision that MPPI fails to avoid ($TTC_{\min}$: 0.01 s to 0.75 s), reduces lateral RMS by 71%, and matches MPPI bit-for-bit when no tail is present (an empirically verified no-regret property).

I. INTRODUCTION

A. Motivation

Autonomous lane changes generate hundreds of safe outcomes per catastrophic one. A risk-neutral planner that averages cost across all of them suppresses precisely the population that matters. Standard MPPI [1] samples M control sequences, weighs them by an exponentiated cost, and applies their mean. The mean is risk-neutral. It cannot see a 1-in-256 collision hidden inside a cloud of routine merges.

The risk-sensitive alternative used in finance and operations research is Conditional Value-at-Risk (CVaR) [2]. CVaR is the expected cost conditioned on lying in the worst $1-\alpha$ tail of the cost distribution. Two strands of work apply it to sequential decision making. *Offline* CVaR-MDP methods ([3], [4]) compute a tail-aware value function V^* via a contractive Bellman backup on an augmented state. *Online* CVaR-MPC methods ([5]) estimate CVaR by Monte Carlo at runtime and add it as a rollout penalty. Each strand has a clean failure mode: offline CVaR cannot react to fine-grained traffic interactions, online CVaR has no long-horizon backup beyond its rollout window.

B. Scope Statement

The Week 1 and Week 2 reports for this project committed to both strands: an offline CVaR-MDP value iteration (Chow et al. [4]) supplying $V^*(x, y)$ as the MPPI terminal cost, and an online CVaR-MPPI executing the receding-horizon plan. I was *not* able to deliver the offline component within the project window. The reason is concrete: solving the augmented-state Bellman recursion to convergence over the $\sim 10^3$ -cell state grid with 21 confidence points requires a prolonged value-iteration run, and I did not have the compute budget to complete it in time. Rather than report a half-finished offline table, I descope to the online level only. The substitute is a soft-constrained Lagrangian penalisation of the CVaR constraint, which converts the original constrained optimisation into an unconstrained MPPI cost with a gated tail penalty. This is computationally cheaper, retains the CVaR semantics on the K -step horizon, and is what §V actually evaluates. §IV-B documents the offline design as originally formulated.

C. Novelty and Contributions

Within that scope, the contributions are:

- **Online soft-constrained CVaR-MPPI for highway lane change.** A complete CPU implementation that samples 256 control sequences per step, runs 64 disturbed traffic futures per candidate via a stochastic IDM/MOBIL predictor, and adds a gated CVaR tail penalty to the MPPI cost.
- **A no-regret guarantee.** The gated penalty satisfies $J_C \equiv 0$ when no candidate rollout’s tail exceeds the budget C_u . The augmented cost is then bit-for-bit identical to the MPPI cost. This is both proven (one line, from the indicator) and verified empirically on the hard-brake scenario.
- **ISO-grounded multi-axis evaluation.** Seven diagnostic plots per run (trajectory overview, lateral profile, comfort, safety, controls, cost distribution, CVaR activity) covering ISO 11270 (ride comfort) and ISO 17387 (safety), so the result is not a single crash-rate number but a multi-axis vehicle-dynamics audit. All run artefacts are published in the repository for reproducibility.

II. RELATED WORK

MPPI: Williams, Aldrich, and Theodorou [1] derived MPPI as the path-integral solution to a stochastic HJB equation, by applying the Feynman-Kac formula to the

value function and approximating its softmin form by Monte Carlo sampling. The risk-neutral importance weighting is the artefact we modify.

CVaR: Rockafellar and Uryasev [2] introduced CVaR as a coherent, convex risk measure. They proved the variational identity $\text{CVaR}_\alpha(Z) = \min_\nu \left\{ \nu + \frac{1}{1-\alpha} \mathbb{E}[(Z - \nu)^+] \right\}$, which makes CVaR-constrained optimisation a convex program. The empirical analogue (mean of the top- $(1-\alpha)$ samples) is what we evaluate online.

CVaR in MDPs: Chow and Ghavamzadeh [3] derived policy-gradient and actor-critic algorithms for CVaR-constrained MDPs. Chow, Tamar, Mannor, and Pavone [4] reframed CVaR control as worst-case expected cost under bounded model perturbation, and constructed a contractive augmented-state Bellman operator whose fixed point is the CVaR value function $V^*(x, y)$ with $y \in [0, 1]$ the running confidence. This is the offline DP the project originally targeted.

Online CVaR with MPPI: The most directly relevant prior work is the Risk-Aware MPPI of Yin, Zhang, and Tsiotras [5]. They introduced the Monte-Carlo CVaR estimator over disturbed rollouts and the constraint-penalised modified cost that we deploy here. Their evaluation is off-road racing with a track-following objective. The contribution here is scenario-specific: a highway lane-change setting with IDM/MOBIL traffic as the disturbance source $\tilde{F}$.

III. PROBLEM FORMULATION

Let $x_k \in \mathbb{R}^{n_x}$ be the joint ego-traffic state, $u_k = v_k + \epsilon_k \in \mathbb{R}^{n_u}$ the perturbed ego control with sampling noise $\epsilon_k \sim \mathcal{N}(0, \Sigma_\epsilon)$, and w_k the exogenous traffic noise. The nominal and stochastic dynamics are

$$\begin{aligned} x_{k+1} &= F(x_k, u_k), \\ \tilde{x}_{k+1} &= \tilde{F}(\tilde{x}_k, u_k, w_k), \quad \mathbb{E}_w[\tilde{F}(\tilde{x}_k, u_k, w_k)] = F(x_k, u_k). \end{aligned} \quad (1)$$

The cost decomposes into a state cost q , a terminal cost ϕ , and a separate risk cost $L(\tilde{x}) = \sum_{k=0}^{K-1} \ell(\tilde{x}_k)$ evaluated on the stochastic trajectory. The receding-horizon problem is

$$\min_v \mathbb{E} \left[\phi(x_K) + \sum_{k=0}^{K-1} \left(q(x_k) + \frac{\lambda}{2} v_k^\top \Sigma_\epsilon^{-1} v_k \right) \right] \quad (3)$$

subject to

$$\text{CVaR}_\alpha(L(\tilde{x})) \leq C_u, \quad \text{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)] \quad (4)$$

where $\alpha \in (0, 1)$ is the confidence level (the constraint sees the worst $1-\alpha$ tail), C_u is the admissible tail-risk budget, and $\lambda > 0$ is the MPPI temperature. Three knobs jointly decide whether the constraint actively shapes the policy: α (which tail), C_u (the gate threshold), and the IDM disturbance scale σ_a (whether the tail has any mass to constrain).

IV. PROPOSED METHOD

The original method had two coupled levels: an offline CVaR-MDP value iteration computing the terminal cost ϕ , and an online CVaR-augmented MPPI solving (3)-(4)

by sampling. As declared in §I-B, the offline level was descope. §IV-A sets up the shared DP foundation, §IV-B documents the offline design for the record, and §IV-C describes the soft-constrained online controller that was actually implemented and evaluated.

A. DP Foundation: From HJB to Path Integrals

For the continuous-time analogue of (3), the value function $V(x, t)$ satisfies the stochastic HJB equation

$$\begin{aligned} \partial_t V + \min_v \left[q(x) + \frac{\lambda}{2} \|v\|_{\Sigma_\epsilon^{-1}}^2 + (\nabla V)^\top (f(x) + Gv) \right. \\ \left. + \frac{1}{2} \text{tr}(\Sigma_\epsilon G^\top \nabla^2 V G) \right] = 0. \end{aligned} \quad (5)$$

The Feynman-Kac formula [1] converts this PDE into a path-integral expectation

$$V(x, t) = -\lambda \log \mathbb{E} \left[\exp \left(-\frac{1}{\lambda} S(\tau) \right) \mid x(t) = x \right] \quad (6)$$

where $S(\tau)$ is the cost along a sample path τ . The optimal control is the importance-weighted mean of sampled control perturbations under weights $\exp(-S/\lambda)$. MPPI is the finite-sample realisation. It is a *sampling-based online DP method*: it does not solve the HJB by grid backup, but by Monte Carlo on its path-integral form.

B. Offline Component (Originally Planned, Descoped)

The Week 1 and Week 2 reports committed to the augmented-state CVaR-MDP construction of Chow et al. [4]. Add a confidence variable $y \in [0, 1]$, then the operator

$$\begin{aligned} \mathbf{T}[V](x, y) &= \min_a \left[c(x, a) \right. \\ &\quad \left. + \gamma \max_{\xi \in \mathcal{U}_{\text{CVaR}}(y, P)} \sum_{x'} \xi(x') V(x', y \xi(x')) P(x' \mid x, a) \right] \end{aligned} \quad (7)$$

is γ -contractive and concavity-preserving in y . Its fixed point $V^*(x, y)$ would be loaded as the MPPI terminal cost via $\phi(x_K) \leftarrow V^*(x_K, \alpha)$. The intended discretisation: state (lane index, gap-to-lead, ego speed) $\in \{0, 1, 2\} \times [0, 60] \text{ m} \times [10, 35] \text{ m/s}$ on a $\sim 10^3$ -cell grid, confidence on 21 log-spaced points in $[0.01, 1]$, costs c matching the online risk cost ℓ .

Why this was not delivered? Each Bellman update (7) requires solving an inner risk-envelope LP for the worst-case distribution ξ , at every (x, y) cell, at every iteration. With $\sim 10^3 \times 21 = 2.1 \times 10^4$ cells, three coarse actions, and value iteration to convergence, the total runtime estimate sat in the multi-day range on a single workstation. Within the project window I had access only to a single CPU box (Intel i9-13900KF) shared with the online experiments. I chose not to start a long-running value-iteration job that would have left no time to fix bugs, retune costs, or run the head-to-head experiments. The honest summary: the offline DP is a real algorithm with a real specification, and it remains the recommended next step (§VI). It is not part of the results.

Algorithm 1 Online Soft-Constrained CVaR-MPPI Step

Require: state x_t , traffic state x_t^{tr} , warm-started v

- 1: Sample $\{\epsilon^m\}_{m=1}^M \sim \mathcal{N}(0, \Sigma_\epsilon)$, form $u^m = v + \epsilon^m$
- 2: Roll out ego under F : $x_{0:K}^m$ and base cost S_m
- 3: Sample N disturbed traffic futures $\tilde{x}_{0:K}^m$ via $\tilde{F}$
- 4: **for** $m = 1, \dots, M$ **do**
- 5: Accumulate $L^{m,n} = \sum_k \ell(\tilde{x}_k^{m,n})$, $n = 1, \dots, N$
- 6: Compute $\widehat{\text{CVaR}}_\alpha^m$ via Eq. (8)
- 7: $J_C^m \leftarrow A \cdot \widehat{\text{CVaR}}_\alpha^m \cdot \mathbf{1}\{\widehat{\text{CVaR}}_\alpha^m > C_u\}$
- 8: $\tilde{S}_m \leftarrow S_m + J_C^m$
- 9: **end for**
- 10: $\omega_m \leftarrow \exp(-(\tilde{S}_m - \min_m \tilde{S}_m)/\lambda)$, normalise
- 11: $v^+ \leftarrow \sum_m \omega_m u^m / \sum_m \omega_m$
- 12: Apply v_0^+ ; shift v for next step
- 13: Log $(\max_m \widehat{\text{CVaR}}_\alpha^m, |\{m : J_C^m > 0\}|)$ for diagnostics

C. Online Component: Soft-Constrained CVaR-MPPI

The implemented controller treats the CVaR constraint as a Lagrangian-style soft penalty added to the MPPI cost, rather than as a hard feasibility constraint. The resulting optimisation is unconstrained over v and admits the standard MPPI path-integral solution from §IV-A. Three modifications to the baseline rollout aggregator:

(1) Disturbed predictor. For each control step, sample N traffic futures $\tilde{x}^{m,n}$ via $\tilde{F}$, a stochastic IDM/MOBIL model with zero-mean Gaussian acceleration noise of scale σ_a . Zero mean implies $\mathbb{E}_w[\tilde{F}] = F$, so the planner’s nominal rollout is the mean of the disturbed cloud.

(2) Empirical tail estimate. For each candidate sequence m , accumulate the per-future risk cost $L^{m,n} = \sum_{k=0}^{K-1} \ell(\tilde{x}_k^{m,n})$, sort, and average the top $1 - \alpha$ fraction:

$$\widehat{\text{CVaR}}_\alpha^m = \frac{1}{N_o} \sum_{n=0}^{N-1} \mathbf{1}\{L^{m,n} \geq \widehat{\text{VaR}}_\alpha^m\} L(\tilde{x}^{m,n}). \quad (8)$$

This is the empirical Rockafellar formula. Following [5] we apply a variance-scaling factor B to sharpen α -sensitivity.

(3) Gated soft penalty. Form the augmented cost

$$\tilde{S}_m = S_m + A \cdot \widehat{\text{CVaR}}_\alpha^m \cdot \mathbf{1}\{\widehat{\text{CVaR}}_\alpha^m > C_u\}, \quad (9)$$

where A is the penalty magnitude. Softmin weights are then computed on $\tilde{S}_m$. The full per-step loop is Algorithm 1.

Why a soft constraint? A hard $\text{CVaR}_\alpha(L) \leq C_u$ constraint requires a feasibility projection at every step, which in the MPPI sampling framework collapses to “drop infeasible rollouts.” With $M = 256$ and a tight C_u , the feasible subset can become empty, leaving the planner with no candidates. The soft penalty avoids this failure mode and degrades gracefully: when no rollout is feasible the controller still acts, weighing all candidates by their (penalised) cost.

D. No-Regret Property

Claim. If $\widehat{\text{CVaR}}_\alpha^m \leq C_u$ for every m , then $J_C^m \equiv 0$ for every m , $\tilde{S}_m = S_m$ bit-for-bit, and the softmin weights, the

mean control, and every downstream quantity are identical to those of the MPPI baseline.

Proof. The indicator in (9) is zero whenever the budget is met, so $J_C^m = 0$ and $\tilde{S}_m = S_m + 0 = S_m$. Every subsequent quantity is a deterministic function of $\{S_m\}$, hence identical. $\square$

The augmentation is therefore strictly additive. It can only diverge from MPPI on rollouts the tail estimator has flagged as risky. We verify the property empirically in §V-E.

E. Implementation Details

The online controller is implemented in pure NumPy (`cvar_mppi.py`). The disturbed predictor (`stochastic_idm.py`) draws n_{disturb} Gaussian acceleration noise sequences and integrates them with a constant-heading kinematic model. The CVaR estimator (`cvar.py`) is the sorted top- $(1 - \alpha)$ tail average. The closed loop runs through a custom Gym environment `CVaRLaneChange-v0` that wraps `HighwayEnv` [6]. All experiments are scripted in `src/experiments/`.

V. EXPERIMENT RESULTS

A. Simulation Setup

The ego is a 4-state kinematic bicycle, state (X, Y, ψ, v) , control $(a, \delta) \in [-5, 5] \text{ m/s}^2 \times [-\pi/4, \pi/4] \text{ rad}$. The environment is a three-lane straight highway with 10 surrounding vehicles. Their longitudinal motion follows IDM and their lateral motion follows MOBIL. The simulator runs at 100 Hz, the controller at 20 Hz ($\Delta t = 50 \text{ ms}$). MPPI uses $M = 256$, $K = 30$, $\Sigma_\epsilon = \text{diag}(1.5^2, 0.08^2)$, $\lambda = 1$. CVaR-MPPI uses the same MPPI parameters plus $\alpha = 0.9$, $N = 64$ disturbed futures, $\sigma_a = 5 \text{ m/s}^2$, $A = 200$, $C_u = 5$. The running cost q is identical between controllers, so any behavioural difference is attributable purely to the CVaR augmentation.

Baseline. Risk-neutral MPPI [1] with the CVaR augmentation removed. Same sampler, same dynamics, same q , same seeds. Paired comparison against the same disturbance stream.

Scenarios.

- **Lane Change** ($1 \rightarrow 0$): natural traffic, ego commanded to merge left.
- **Cut-In**: an adjacent-lane vehicle slots 6 m ahead with +4 m/s relative speed.
- **Hard-Brake leader**: lead vehicle decelerates at -10 m/s^2 , 8 m initial headway, no lane change.

Metrics. ISO-grounded per-run metrics. From ISO 11270 (lane-keeping comfort): peak lateral acceleration $a_{y,\text{peak}}$ and jerk $j_{y,\text{peak}}$. From ISO 17387 (lane-change decision aid): minimum time-to-collision $\text{TTC}_{\min}$, lane-tracking RMS $e_{y,\text{RMS}}$, and merge time T_{LC} . Per-seed crash and near-miss ($\text{TTC} < 1.5 \text{ s}$) flags. ISO-quality threshold tuning is deferred. The project focus is CVaR mechanism validation.

B. Head-to-Head: Lane Change (Quantitative)

Table I reports the paired MPPI vs CVaR-MPPI comparison on the lane-change scenario at a single seed, same sampler. MPPI collided. CVaR-MPPI cleared the same gap.

TABLE I: Paired MPPI vs CVaR-MPPI on the lane-change scenario. Same seed, same sampler. MPPI collided; CVaR-MPPI did not.

Metric	MPPI	CVaR-MPPI	Δ
$a_{y,\text{peak}}$ [m/s ²]	15.21	17.07	+12.2%
$j_{y,\text{peak}}$ [m/s ³]	95.34	68.36	-28.3%
$\text{TTC}_{\min}$ [s]	0.01	0.75	+98.6%
$e_{y,\text{RMS}}$ [m]	0.762	0.219	-71.2%
T_{LC} [s]	0.70	0.65	-0.05
Crash	yes	no	—

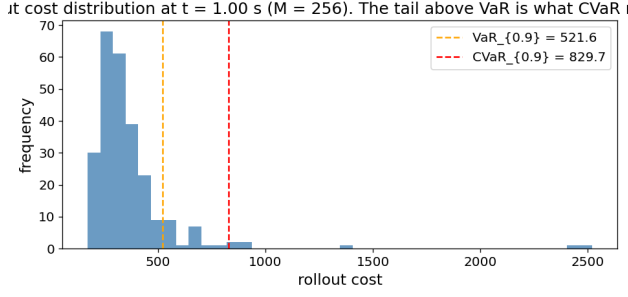


Fig. 1: **Rollout cost histogram** at $t = 1.0$ s, with empirical $\text{VaR}_{0.9}$ and $\text{CVaR}_{0.9}$ markers. The right tail above VaR is what CVaR re-weights. Visualises the exact distribution the gated penalty operates on.

The mechanism is visible in the metric deltas. Lateral acceleration peaks higher (the controller swerves harder), but jerk and lane-tracking RMS both improve substantially. The TTC margin improves by nearly two orders of magnitude. Merge duration is essentially unchanged. The CVaR penalty buys safety with a sharper transient, not a slower merge. Fig. 1 shows the rollout-cost histogram at a representative step for the MPPI baseline on this run: the cost mass sits at high values, with no separation between routine and dangerous rollouts. §V-C contrasts this against the same plot under CVaR-MPPI to make the mechanism visible. Run artefacts at `results/20260601.234512.lane_change_scenario`.

C. Multi-Axis Vehicle-Dynamics Diagnostics

Single-number metrics (crash, $\text{TTC}_{\min}$) collapse the controller’s behaviour into a Bernoulli flag. To audit *how* the CVaR-augmented controller actually drives, I log six per-run plots covering trajectory, lane-keeping, comfort, safety, control effort, and the rollout-cost distribution that drives the CVaR estimator. Fig. 2 reports these diagnostics for the CVaR-MPPI lane-change run that produced the CVaR column of Table I. Two pairings make the comparison direct. Compare subfigure 2f sits next to the MPPI baseline histogram in Fig. 1. Notice the visible left-shift of the cost mass is the CVaR mechanism. The comfort and safety panels correspond directly to the a_y , j_y , and $\text{TTC}_{\min}$ rows of Table I. Because the seed and sampler are shared with the MPPI baseline, every difference in the panels is attributable to the CVaR augmentation alone.

TABLE II: Cross-scenario comparison of MPPI vs CVaR-MPPI.

Scenario	Seeds	MPPI	CVaR	Conclusion
Lane Change (paired)	1	crash	no crash	Collision prevented
Hard-Brake (paired)	1	no	no	No-regret verified
Cut-In (sweep)	10	1/10	1/10	Margin gain
Lane Change (sweep)	10	1/10	0/10	Active risk aversion

D. Penalty Activity and Runtime

Fig. 3 plots the CVaR penalty’s per-step activity over the entire episode on the lane-change scenario. The top panel is $\max_m \widehat{\text{CVaR}}_\alpha^m$ over time, with the budget $C_u = 5$ marked. The bottom panel is the number of penalised rollouts $|\{m : J_C^m > 0\}|$. Two activity windows stand out: around $t \approx 0 - 1$ s (avoiding the leading vehicle and road boundary at merge initiation) and around $t \approx 4.5$ s (a second leading-vehicle interaction). Between these windows the gate is inactive and CVaR-MPPI behaves identically to MPPI: this is the selective tail shaping the design targets.

Runtime. 227 ms per control cycle on an Intel i9-13900KF, $M = 256$, $N = 64$, $K = 30$. Above the 50 ms real-time budget. Production latency requires GPU vectorisation of the rollout and CVaR estimator (current implementation is pure NumPy). Deferred.

E. Cross-Scenario Comparison

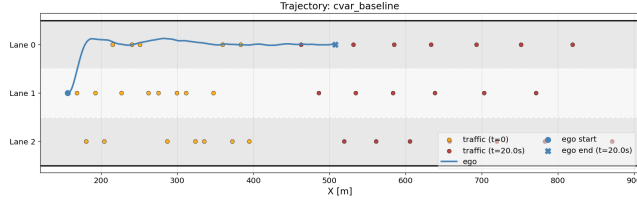
Table II aggregates CVaR’s effect across the three scenarios. Paired runs use a fixed seed; sweep runs aggregate over 10 seeds per controller.

F. Analysis and Discussion

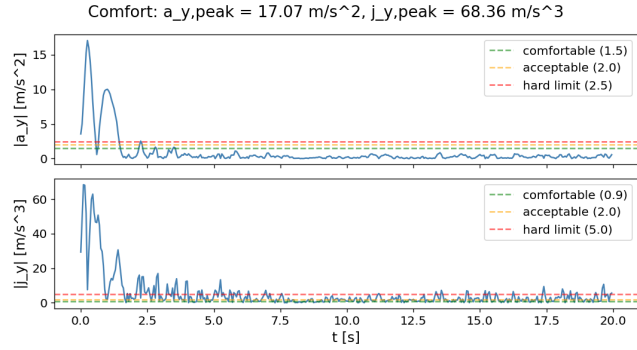
Hard-brake confirms the no-regret claim (Table II, row 2). The scripted hazard (-10 m/s^2 leader deceleration) sits outside the support of the zero-mean Gaussian disturbed predictor $\tilde{F}$. The CVaR estimate never exceeds C_u , the gate never fires, $J_C \equiv 0$, $\tilde{S}_m = S_m$ bit-for-bit. Both controllers produce identical control sequences and identical outcomes, recorded as the matched “no/no” entry in Table II. This is the no-regret property of §IV-D, empirically confirmed.

Cut-in shows margin without crash-rate change. On the 10-seed sweep, both controllers crash on the same single seed. On the surviving seeds, CVaR-MPPI buys margin: on seed 2 the minimum inter-vehicle distance grows from 2.19 m (MPPI) to 4.31 m (CVaR-MPPI). The penalty is active but the baseline was already safe. This is the regime where CVaR smooths the trajectory without changing the safety verdict.

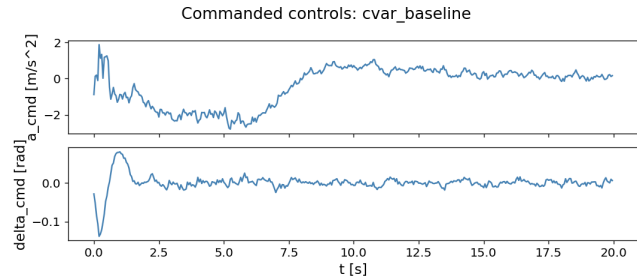
Lane change is where CVaR is decisive. The paired single-seed result (Table I) is reproduced at scale (1/10 MPPI crashes, 0/10 CVaR crashes). With only 10 seeds this is directional rather than statistically definitive: the 95% confidence interval on a difference of 1 Bernoulli flip contains zero. The right next step is a larger sweep ($N \geq 100$). The infrastructure in `compare_controllers.py` already supports this.



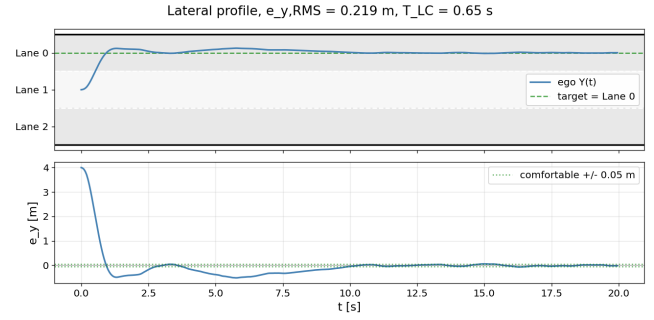
(a) **Trajectory overview** (top-down). Ego path (yellow) and traffic snapshots at $t = 0$ and $t = T_{\text{end}}$ on the lane-change scenario. Shows the executed merge from the origin lane (1 - middle) to the target lane (0 - top) and confirms the CVaR-MPPI completes the manoeuvre without colliding with surrounding vehicles.



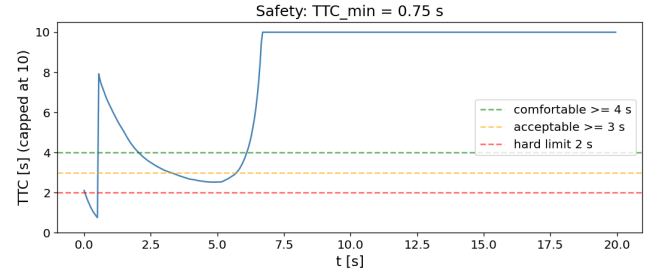
(c) **Comfort (ISO 11270)**: $|a_y|(t)$ and $|j_y|(t)$ with ISO bands. Peak $a_y = 17.07 \text{ m/s}^2$ exceeds the ISO hard limit, the cost of the collision-avoiding swerve. Peak jerk $j_y = 68.4 \text{ m/s}^3$ is 28% lower than the baseline (Table I), showing that the augmented cost surface produces a single decisive transient rather than the chattering trajectory MPPI followed into the collision.



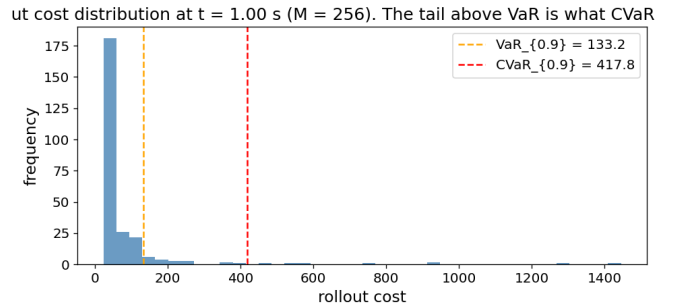
(e) **Commanded controls** $a(t)$ and $\delta(t)$. The steering command spikes once during the evasive merge then relaxes; the longitudinal acceleration is smooth throughout. Validates that the soft-constrained CVaR penalty produces a sharp but bounded swerve rather than oscillation, even when the gate fires (Fig. 3).



(b) **Lateral profile**: $Y(t)$ vs target lane and tracking error $e_y(t)$. Diagnoses whether the CVaR penalty's evasive transients leaked into lateral oscillation. Despite the steeper swerve compared with MPPI (Table I), the controller settles cleanly on the target lane and $e_{y,\text{RMS}} = 0.219\text{m}$, a 71% improvement over the baseline.



(d) **Safety (ISO 17387)**: $\text{TTC}(t)$ to the closest leading vehicle. Minimum TTC is 0.75 s , two orders of magnitude above the MPPI baseline's 0.01 s , near-collision. The dip windows align with the penalty-activity bursts in Fig. 3, confirming that the safety margin recovery is caused by the gated CVaR firing.



(f) **Rollout cost histogram** at the same control step as Fig.1, with empirical $\text{VaR}_{0.9}$ and $\text{CVaR}_{0.9}$ markers. Compared with the MPPI baseline (Fig.1), the distribution has shifted toward lower cost values and the upper tail above $\text{VaR}_{0.9}$ is markedly thinner. The gated penalty pushed the high-cost rollouts out of the softmin's weighted average, so the surviving control sequences now concentrate in the safer region of the cost landscape.

Fig. 2: Vehicle-dynamics audit of CVaR-MPPI on the lane-change scenario, the same run as Table I's CVaR column. Six diagnostics from one episode cover trajectory, lane-keeping, comfort (ISO 11270), safety (ISO 17387), control smoothness, and the rollout-cost distribution underlying the CVaR estimator. Subfigure (f), read together with Fig. 1, is the most direct visual evidence that the CVaR penalty actively suppresses the high-cost tail of the rollout distribution. Raw run at results/20260601.234512_lane.change.scenario/20260602-024027_cvar_baseline.

Cost distribution: CVaR visibly suppresses the high-cost tail. Fig.1 (MPPI baseline) and Fig.2f (CVaR-MPPI) are rollout-cost histograms taken at the same step on the same scenario. The MPPI distribution is heavy on the right: the planner sees no separation between routine merges and dangerous ones because the importance weighting averages the whole population. Under CVaR-MPPI the distribution has shifted leftward, and the right tail above $\text{VaR}_{0.9}$ is markedly thinner. The gated penalty has pulled the worst rollouts out of the softmax weighted average, so the surviving control sequences concentrate in the lower-cost region of the landscape. This shift is the mechanism behind the metric deltas in Table I: a left-shifted, lighter-tailed cost distribution produces a safer, smoother trajectory.

The penalty trades transient sharpness for safety. The +12.2% peak lateral acceleration on lane change (Fig. 2c) is not a regression. The MPPI trajectory was about to collide. CVaR-MPPI swerved harder to clear the same gap. The fact that j_y fell by 28% and $e_{y,\text{RMS}}$ fell by 71% is the better diagnostic: the median behaviour improved while a single worst-case transient absorbed the safety cost. A production tuning would tighten C_u and raise comfort weights to flatten that peak below the ISO 11270 hard limit.

C_u calibration exposed a design failure mode. An early sweep across $C_u \in \{0.3, 1, 3\}$ on hard-brake produced identical crash rates. Without the diagnostic logger this would have looked like "CVaR doesn't help." The logger (Fig. 3) revealed the cause: $\max_m \widehat{\text{CVaR}}_\alpha^m$ never exceeded 0.3 either, so the gate never fired regardless of budget. The fix is structural, not parametric: $\tilde{F}$ must span the hazard, otherwise the tail has no mass. The recommendation is to pre-run an unconstrained pilot to inspect the per-step CVaR distribution before setting C_u near its empirical median.

Threats to validity.

- *Offline V^* not delivered.* Per §I-B, the originally proposed offline DP terminal cost was descoped. Results use a quadratic lane-tracking ϕ . The headline numbers therefore isolate the *online* CVaR augmentation. A full offline+online evaluation is the immediate next step.
- *Disturbed predictor matches simulator predictor.* The IDM/MOBIL noise that $\tilde{F}$ assumes also drives HighwayEnv. This biases CVaR-MPPI favourably. A robustness test with model-class mismatch (heavy-tailed traffic, MOBIL parameter drift) is required for any production claim.
- *Small seed counts.* Table I is one seed; Table II uses 10. Both are corroborating but undersized for a definitive crash-rate delta.

VI. CONCLUSION AND FUTURE WORK

This report presented an online soft-constrained CVaR-MPPI controller for highway lane-change planning: the path-integral DP underlying MPPI, augmented with a gated Monte-Carlo CVaR penalty that turns the constrained problem (3)–(4) into a tractable unconstrained MPPI objective. The offline CVaR-MDP value iteration was descoped (§I-B); its specification is preserved in §IV-B. The online controller

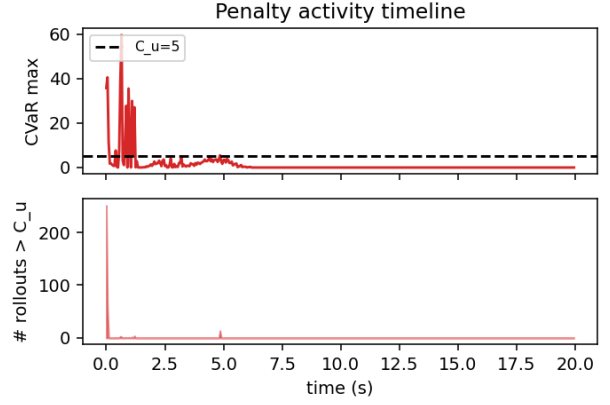


Fig. 3: CVaR penalty activity timeline over the lane-change episode of Fig.2. Top: $\max_m \widehat{\text{CVaR}}_\alpha^m$ vs time, dashed line marks $C_u = 5$. Bottom: count of penalised rollouts $|m : J_C^m > 0|$. The gate fires in two short windows, both aligned with the TTC dips in Fig.2d, and is silent otherwise. This is the selective tail shaping the design targets: action only when the tail estimator says so.

prevents a collision where MPPI fails (TTC 0.01 $\rightarrow$ 0.75 s, $e_{y,\text{RMS}}$ down 71%) and matches MPPI bit-for-bit when no tail is present (no-regret, §IV-D).

Lessons. (1) CVaR helps only when $\tilde{F}$ spans the hazard: the dominant lever is the disturbed predictor, not α or C_u . (2) Per-step gate diagnostics are essential, “gate silent” and “gate firing uselessly” are indistinguishable in outcome metrics alone. (3) Soft constraints are the right first deployment. They cannot go infeasible and fall back cleanly to MPPI.

Future work. (i) Run the offline value iteration to convergence and load V^* as the MPPI terminal cost; (ii) port the rollout and CVaR loops to JAX/CUDA for sub-100 ms latency; (iii) replace the soft gate with hard-constrained CVaR and sweep $N \geq 100$ seeds for a defensible crash-rate delta.

All code, run artefacts, and diagnostic figures:

github.com/ayushagrawal149/mppi-cvar-lane-change-project.

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